

A partial sharp Rademacher case for the noncommutative Khintchine constant

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Status

This is a partial result for the open sharp-constant question in Haagerup–Musat’s paper [1]. The general Rademacher case remains open. The result below proves that the conjectural lower constant $1/\sqrt{2}$ is valid for all commuting normal coefficient families and for arbitrary two-term matrix coefficient families.

Source evidence

Haagerup–Musat prove the Rademacher version with lower constant $1/\sqrt{3}$, and state that they do not know whether this is sharp. The relevant passage is on page 2 of [1].

where, by definition,

$$(1.3) \quad |||\{x_i\}_{i=1}^d|||_* : = \inf \left\{ \text{Tr} \left(\left(\sum_{i=1}^d y_i^* y_i \right)^{\frac{1}{2}} + \left(\sum_{i=1}^d z_i z_i^* \right)^{\frac{1}{2}} \right); x_i = y_i + z_i \in M_n(\mathbb{C}) \right\}.$$

Here S_1^n is $M_n(\mathbb{C})$ with the norm $\|x\|_1 := \text{Tr}((x^*x)^{1/2})$, and Tr is the non-normalized trace on $M_n(\mathbb{C})$. We should also point out that it was noted in the paper [14] (cf. p. 250) that, by using the lacunary sequence $\{3^n\}_{n \geq 1}$ instead of the sequence $\{2^n\}_{n \geq 1}$, the lower bound in the inequality (1.2) can be improved to $\frac{1}{2}$.

By classical arguments (cf. Proposition 3.2 in [18]), if one replaces $\{e^{i2^n t}\}_{n=1}^\infty$ by a sequence of independent complex Gaussian, respectively, Rademacher or Steinhaus random variables, the corresponding Khintchine inequality with matrix coefficients follows, as well, with possibly different constants.

Our method, leading to improved constants, was inspired by ideas of Pisier from [17], and it is based on proving first directly the dual inequality to (1.2) with constant $\sqrt{2}$, where $\{e^{i2^n t}\}_{n=1}^\infty$ is replaced by a sequence of independent complex-valued standard Gaussian random variables on some probability space $(\Omega, \mathbb{P})$. Based on a result from [9], the constant $\sqrt{2}$ turns out to be optimal in this case, and for the sequence $\{e^{i2^n t}\}_{n=1}^\infty$. We also consider the case of a sequence of Rademacher functions, and prove that the corresponding noncommutative Khintchine inequality holds with constant $\sqrt{3}$ instead of $\sqrt{2}$, but we do not know yet whether this is sharp.

In the second part of our paper we obtain an improvement of a recent result of M. Junge (cf. [11]) concerning a Khintchine-type inequality for subspaces of $R \oplus_\infty C$ (the l^∞ -sum of the row and column Hilbert spaces). Recall that $R := \text{Span}\{e_{1j}; j \geq 1\}$, respectively, $C := \text{Span}\{e_{j1}; j \geq 1\}$, where e_{kl} is the element in $\mathcal{B}(\ell_2)$ corresponding to the matrix with entries equal to 1 on the (k, l) position, and 0 elsewhere. This Khintchine-type inequality is intimately connected with the question of the existence of a completely isomorphic embedding of the operator space OH , introduced by G. Pisier (see [19]), into a noncommutative L^1 -space, a problem that was resolved by Junge in the remarkable paper [10]. In [11] (see Section 8), Junge improved this result, by showing that OH cb-embeds into the predual of a hyperfinite type III_1 factor.

In our new approach, we first observe that given a closed subspace H of $R \oplus C$, there is a self-adjoint operator $A \in \mathcal{B}(H)$ satisfying $0 \leq A \leq I$, where I denotes the identity operator on H , such that the operator space structure on H is given by

$$(1.4) \quad \left\| \sum_{i=1}^r x_i \otimes \xi_i \right\|_{M_n(H)} = \max \left\{ \left\| \sum_{i,j=1}^r \langle (I - A)\xi_i, \xi_j \rangle_H x_i x_j^* \right\|^{1/2}, \left\| \sum_{i,j=1}^r \langle A\xi_i, \xi_j \rangle_H x_i^* x_j \right\|^{1/2} \right\},$$

where n, r are positive integers, $x_1, \dots, x_r \in M_n(\mathbb{C})$ and $\xi_1, \dots, \xi_r \in H$.

As in Junge's approach from [11], we will use CAR algebra methods. We consider the associated quasi-free state ω_A on the CAR-algebra $\mathcal{A} = \mathcal{A}(H)$ built on the Hilbert space H , and construct a linear map F_A of H^* into the predual M_* of the von Neumann algebra $M := \overline{\pi_A(\mathcal{A})}^{\text{wot}}$, which by [23] is a hyperfinite factor. Here π_A is the unital $*$ -homomorphism from the GNS representation associated to $(\mathcal{A}, \omega_A)$. Note that M_* can be considered as a subspace of $\mathcal{A}^*$. Next we let F_A be the transpose of the map $E_A : \mathcal{A} \rightarrow H$ defined by

$$(1.5) \quad \langle E_A(b), f \rangle_H = \omega_A(ba(f)^* + a(f)^*b), \quad \forall b \in \mathcal{A}, \forall f \in H,$$

Source problem

For $x_1, \dots, x_d \in M_n(\mathbb{C})$, set

$$N(x_1, \dots, x_d) = \inf_{x_i = y_i + z_i} \operatorname{Tr} \left(\sum_i y_i^* y_i \right)^{1/2} + \operatorname{Tr} \left(\sum_i z_i z_i^* \right)^{1/2}.$$

Haagerup–Musat prove

$$\frac{1}{\sqrt{3}} N(x_1, \dots, x_d) \leq \mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i x_i \right\|_{S_1^n} \leq N(x_1, \dots, x_d),$$

where (ε_i) are independent Rademacher signs. They also record the scalar obstruction showing that the best possible lower constant is at most $1/\sqrt{2}$. The open sharpness issue is whether the lower constant can be improved from $1/\sqrt{3}$ to $1/\sqrt{2}$ in full generality.

Proof intuition

The two subcases below remove different sources of noncommutativity. For commuting normal matrices, simultaneous diagonalization reduces both the trace norm and the N -norm to pointwise scalar quantities; Szarek’s sharp scalar $p = 1$ Khintchine inequality then applies on each diagonal fiber. For two arbitrary matrices, a 2×2 Hadamard rotation of the row block $[x_1 \ x_2]$ gives the needed comparison with $\|x_1 + x_2\|_1 + \|x_1 - x_2\|_1$, which is exactly twice the Rademacher average. Thus the remaining obstruction, if any, must use at least three noncommuting coefficients.

Lemma 1 (Diagonal formula). *Suppose $x_i = \operatorname{diag}(a_i(1), \dots, a_i(n))$ are diagonal matrices. Then*

$$N(x_1, \dots, x_d) = \sum_{k=1}^n \left(\sum_{i=1}^d |a_i(k)|^2 \right)^{1/2}.$$

Proof. The upper bound follows by taking $y_i = x_i$, $z_i = 0$. For the lower bound, let $D : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ be the diagonal conditional expectation. It is the average of conjugations by diagonal sign unitaries. Hence the maps

$$(u_i)_i \mapsto (Du_i)_i$$

are contractions for both norms

$$(u_i)_i \mapsto \operatorname{Tr} \left(\sum_i u_i^* u_i \right)^{1/2}, \quad (u_i)_i \mapsto \operatorname{Tr} \left(\sum_i u_i u_i^* \right)^{1/2}.$$

Therefore any decomposition $x_i = y_i + z_i$ can be replaced by the diagonal decomposition $x_i = Dy_i + Dz_i$ without increasing the objective. Once all matrices are diagonal, the objective is

$$\sum_{k=1}^n (\|b(k)\|_2 + \|c(k)\|_2), \quad b(k) + c(k) = a(k),$$

where $a(k) = (a_i(k))_i$. The pointwise infimum is exactly $\|a(k)\|_2$, proving the formula. $\square$

Theorem 1 (Sharp abelian subcase). *If $x_1, \dots, x_d \in M_n(\mathbb{C})$ are commuting normal matrices, then*

$$\mathbb{E}_\varepsilon \left\| \sum_{i=1}^d \varepsilon_i x_i \right\|_{S_1^n} \geq \frac{1}{\sqrt{2}} N(x_1, \dots, x_d).$$

The constant $1/\sqrt{2}$ is best possible even in this abelian class.

Proof. After unitary conjugacy the x_i 's are diagonal, $x_i = \text{diag}(a_i(1), \dots, a_i(n))$. By Lemma 1,

$$N(x_1, \dots, x_d) = \sum_{k=1}^n \|a(k)\|_2.$$

On the other hand,

$$\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i x_i \right\|_1 = \sum_{k=1}^n \mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i a_i(k) \right|.$$

Szarek's sharp scalar $p = 1$ Khintchine lower bound gives

$$\mathbb{E}_\varepsilon \left| \sum_i \varepsilon_i a_i(k) \right| \geq \frac{1}{\sqrt{2}} \|a(k)\|_2$$

for every k . Summing over k proves the asserted inequality.

Sharpness follows from the scalar two-term choice $n = 1$, $x_1 = x_2 = 1$, for which $N(x_1, x_2) = \sqrt{2}$ and $\mathbb{E}|\varepsilon_1 + \varepsilon_2| = 1$. $\square$

Theorem 2 (Sharp arbitrary two-term subcase). *For all $x_1, x_2 \in M_n(\mathbb{C})$,*

$$\mathbb{E}_{\varepsilon_1, \varepsilon_2} \|\varepsilon_1 x_1 + \varepsilon_2 x_2\|_{S_1^n} \geq \frac{1}{\sqrt{2}} N(x_1, x_2).$$

The constant $1/\sqrt{2}$ is best possible.

Proof. Since the trace norm is invariant under multiplication by -1 ,

$$\mathbb{E}_{\varepsilon_1, \varepsilon_2} \|\varepsilon_1 x_1 + \varepsilon_2 x_2\|_1 = \frac{1}{2} (\|x_1 + x_2\|_1 + \|x_1 - x_2\|_1).$$

Let $R = [x_1 \ x_2]$ be the row block operator. Then

$$\|R\|_{S_1} = \text{Tr}(x_1 x_1^* + x_2 x_2^*)^{1/2}.$$

Right multiplication by the 2×2 Hadamard unitary gives

$$\|R\|_{S_1} = \frac{1}{\sqrt{2}} \|[x_1 + x_2 \quad x_1 - x_2]\|_{S_1} \leq \frac{1}{\sqrt{2}} (\|x_1 + x_2\|_1 + \|x_1 - x_2\|_1).$$

In the definition of N , taking $y_1 = y_2 = 0$ and $z_i = x_i$ gives

$$N(x_1, x_2) \leq \text{Tr}(x_1 x_1^* + x_2 x_2^*)^{1/2} = \|R\|_{S_1}.$$

Combining the last three displays yields

$$N(x_1, x_2) \leq \sqrt{2} \mathbb{E}_{\varepsilon_1, \varepsilon_2} \|\varepsilon_1 x_1 + \varepsilon_2 x_2\|_1,$$

which is the claim. The scalar example $x_1 = x_2 = 1$ again gives equality. $\square$

Verification notes

The proof is deterministic. The included script `code/verify_subcases.py` numerically checks the diagonal norm formula and the two-term row-block estimate on random small matrices. These checks are only sanity tests, not part of the proof.

Limitations and novelty check

This is not a full solution of the Haagerup–Musat Rademacher sharpness problem. It leaves open the case of three or more genuinely noncommuting coefficients. A local corpus search found Pisier’s 2011 survey text still recording the Rademacher best constants as apparently unknown [3]. A 2025 preprint by Chuah–Liu–Mei also states that the exact Rademacher operator-valued value is still unknown while proving related Z_2 -set criteria [4]. No exact literature packet for arXiv:math/0611160 was present in the run registry at promotion time.

Human-review recommendation

Check Lemma 1, especially the use of the diagonal conditional expectation as a contraction on the two $S_1(\ell_2)$ row/column norms. The two-term argument is a short block-matrix calculation and should be straightforward to verify.

References

- [1] U. Haagerup and M. Musat, *On the best constants in noncommutative Khintchine-type inequalities*, J. Funct. Anal. 250 (2007), 588–624; arXiv:math/0611160.
- [2] S. J. Szarek, *On the best constants in the Khinchin inequality*, Studia Math. 58 (1976), 197–208.
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- [4] C. Y. Chuah, Z.-C. Liu, and T. Mei, *Noncommutative Khintchine inequalities and Z_2 property*, arXiv:2503.03187.